



CASE STUDY OF CYBER PHYSICAL PRODUCTION SYSTEM USING AM

INVOLUTE GEARING : INFLUENCE OF TOOTH HEIGHT

GROUP NUMBER : 13

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Abstract:

The technical report investigates the effects of changing the tooth height in involute gear. The effects of various addendum adjustments on the contact ratio and stress concentrations in gear teeth are investigated numerically. According to the results, raising the addendum raises the contact ratio, which could lead to better load distribution and operational effectiveness. But this also causes more tension at the tooth root, which could result in an early gear failure. Lowering the addendum raises the stress per unit area and increases wear since it lowers the contact ratio. The study employs three gear set configurations to assess the effect of addendum modification using numerical formulations. Hertz Contact stress equation is used to compare the contact stresses in the gear pairs. The study describes the numerical techniques applied to evaluate these impacts and offers helpful suggestions for refining gear tooth design to strike a compromise between performance and durability.

Keywords: Involute gear, gear geometry, addendum modifications, Hertzian Contact Stress, Profile shift

Abstrakt:

Der technische Bericht untersucht die Auswirkungen der Änderung der Zahnhöhe in involute Gear. Numerisch werden die Auswirkungen verschiedener Addendumanpassungen auf die Kontaktverhältnisse und Stresskonzentrationen in Gear-Teeth untersucht. Nach den Ergebnissen erhöht die Erhöhung des Addendums den Kontaktverhältnis, was zu einer besseren Verteilung der Belastung und zu einer höheren operativen Wirksamkeit führen könnte. Das führt jedoch zu einer erhöhten Spannung an der Zahnwurzel, was zu einem frühzeitigen Gear-Ausfall führen kann. Lowering the addendum erhöht den Stress pro Unit Area und verursacht eine Erhöhung der Wear, da es den Kontaktverhältnis senkt. Die Studie verwendet drei Gear-Set-Konfigurationen, um den Effekt der Addendum-Modifikation mit numerischen Formulierungen zu bewerten. Die Hertz Contact Stress Equation wird verwendet, um die Kontaktstresse in den Gear-Pairs zu vergleichen. In der Studie werden die numerischen Methoden beschrieben, mit denen diese Auswirkungen bewertet wurden, und es werden nützliche Vorschläge für die Verbesserung des Zahndesigns von Gear gemacht, um eine Balance zwischen Leistung und Langlebigkeit zu finden.

Schlüsselwörter: Involute Gear, Gear Geometrie, Addendum-Änderungen, Hertzian Kontakt Stress, Profilwechsel

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1 Introduction

Involute gears are a distinctive tooth profile, that is used in automotive to industrial machineries. Understanding the dynamics of contact points between meshing gear teeth is important. The contact point is where force transmission between gear teeth takes place, and it has a significant impact on the gear's longevity, noise level, and efficiency. Precisely locating these points aids in anticipating complications that might lead to tooth fracture or pitting. Pressure angles, addendum adjustments, and the tooth profile all affect the distribution of Hertzian contact stress. The tooth profile can be adjusted by adjusting the addendum height, affecting the gear's longevity and performance. In positive addendum modification, the base profile is shifted outwards, whereas, in negative addendum modification the base profile is shifted inwards [4] [9].

Shuting Li [7] used the mathematical method, and finite element method together to calculate the loaded tooth contact analyses, deformation, and stress of spur gears with different addendum and contact ratios. With an emphasis on space applications, Wood. J New. G [12] studied design and manufacturing issues for modified involute gearing. Depending on the gear train's requirements, various levels of modification, including profile modification, may be required to maintain a long life and smooth operation. Nayak. S & Mishra. Using mathematical formulation and finite element analysis method, S used a range of addendum modification coefficient values from positive to negative to indicate the influence of addendum on root stresses. Atanasiu. V et. al [2] analyzed the relation between the contact pressure of meshing teeth and specific sliding with addendum modification coefficients. Damtie. S & Tilahun. D [5] designed the involute gear in SolidWorks and compared the contact stress using the finite element analysis in the Ansys and Hertzian formula. Babu. S & M. A. [3] utilized a new approach to design to gear design which integrates the simplicity of Excel spreadsheets with the advanced graphical capabilities of SolidWorks. Hassan. A [6] presented a pair of mating gear during rotation and analyzed contact stress in different contact positions.

This study illustrates the impact of tooth height in involute spur gears using three distinct gear configurations of gear sets. Standard gear is taken into account in all three configurations, while the addendum modification coefficients of $+0.25$, 0 and -0.25 in pinion vary the gear sets. The results

will be calculated using the gear geometries formula. Hertzian Contact stress is used to analyze and compare the three different gear pairs.

2 Involute Gear Geometry and its equation

2.1 Involute curve & its parametric equation:

The involute curve is the path drawn by unwinding a string from its base circle by keeping it tight and letting the end trace out a curve. The involute curve is based on the radius (r_b) and hence the parametric equations are:

$$x(\theta) = r_b(\cos(\theta) + (\theta)\sin(\theta))$$

$$y(\theta) = r_b(\sin(\theta) - (\theta)\cos(\theta))$$

Here, $x(\theta)$ & $y(\theta)$ are the coordinates on the involute curve, r_b is the base radius and θ is the angle parameter which corresponds to the angle by which the string has been unwound from the base circle.

2.2 Involute angle calculation:

An additional method to represent an involute curve is by using a variable parameter θ , where θ is the rolling angle. The derivation of the involute curve may be accomplished as follows [8]:

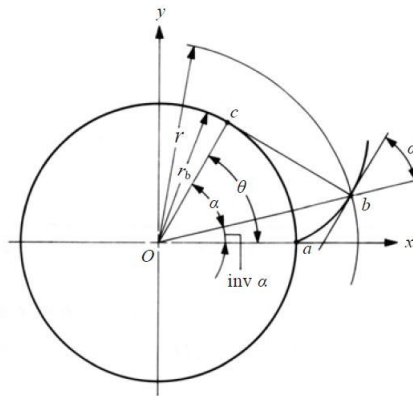


Figure 1: Involute Curve [1]

$$x = r \sin \theta$$

$$y = r \cos \theta$$

$$r = \frac{r_b}{\cos \alpha}$$

$$\theta = \tan \alpha - \alpha$$

Function $\theta(\alpha)$ is designated as $\text{inv}(\alpha)$, hence the parametric equations become:

$$x = \frac{rb}{\cos \alpha} \sin (\text{inv}(\alpha)) \quad y = \frac{rb}{\cos \alpha} \cos (\text{inv}(\alpha))$$

Function

$$\text{inv}(\alpha) = \tan \alpha - \alpha$$

2.3 Standard involute gear parameters with the basic formula:

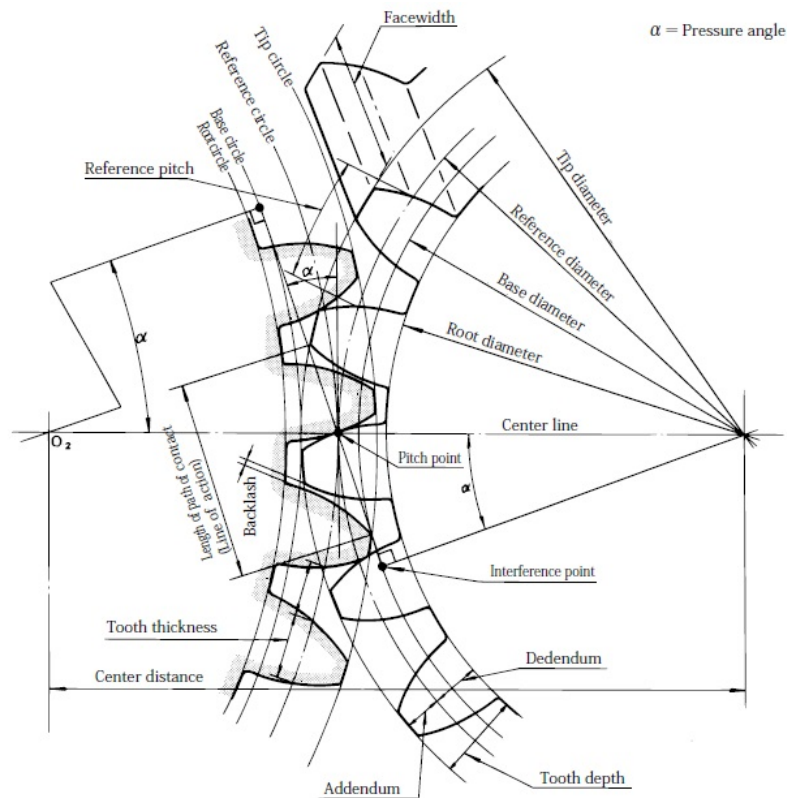


Figure 2: Gear Terminologies [1]

Module (m): Module is the foundational concept in the metric system for gear specification, and it is a measure of the size of the gear teeth.

Pressure angle (α): The pitch circle tangent from which the teeth of the gear are inclined is identified as the pressure angle.

Pitch circle diameter (d): It is the imaginary circle that passes through the point where the teeth of two gears effectively mesh.

$$d = m \times z$$

Base circle diameter (d_b): The diameter of the base circle is the circle from which the involute curve, a geometric shape that forms the tooth profile, originates.

$$d_b = d \times \cos(\alpha)$$

Base pitch (p_b): The base pitch can be expressed mathematically as.

$$p_b = m\pi \times \cos(\alpha)$$

Addendum coefficient (h_a^*): Addendum coefficient is a dimensionless parameter that defines the addendum. For standard gears, the addendum coefficient is 1.

$$h_a^* = 1$$

Dedendum coefficient (h_f^*): Addendum coefficient is a dimensionless parameter that defines the dedendum. For standard gears, the dedendum coefficient is 1.25.

$$h_f^* = 1.25$$

Addendum height (h_a): The radial distance of the tooth from the pitch circle to the top of the tooth.

$$h_a = m \times h_a^*$$

Dedendum height (h_f): The radial distance of the tooth from the pitch circle to the bottom of the tooth.

$$h_f = m \times h_f^*$$

Tip circle (d_a): The tip circle is the circle drawn through the top of the tooth.

$$d_a = d + 2m$$

Root circle (d_f): This is the circle that bounds the base of the gear teeth.

$$d_f = d - 2.5m$$

Clearance (c): Clearance is the radial distance between the top of the tooth of one gear to the bottom of the tooth space on the mating gear.

$$c = a - \frac{d_{f1}}{2} - \frac{d_{a2}}{2}$$

Where, d_{f1} is the dedendum radius of pinion and d_{a2} is the addendum radius of gear.

Center distance (a): It is the distance between the centers of the two meshing gears.

$$a = \frac{d_1 + d_2}{2}$$

Where, d_1 & d_2 are the pitch circle of pinion and gear.

Tooth thickness (s): The width of the gear tooth measured along the pitch circle is tooth thickness.

$$s = \frac{\pi m}{2}$$

Tooth space (e): The distance along the pitch circle between neighboring teeth is called tooth space.

$$e = \frac{\pi m}{2}$$

3 Addendum modification and modified gear formulas

3.1 Addendum modification coefficient (x):

One major issue with gears with few teeth is interference. Accordingly, "undercutting" is carried out to prevent interference in the gears. Yet, the tooth becomes weaker due to undercutting. Still, there are instances where gears will need to function with fewer teeth than the minimal number specified to prevent undercutting. It's referred to as the Addendum Modification in gear technology. The profile shift and addendum modification are the same. The formula to calculate the addendum modification coefficient is [9]

$$x = 1 - z \frac{\sin^2(\alpha)}{2}$$

The term $\frac{(\sin^2(\alpha))}{2}$, is the reciprocal of the minimum number of teeth. For a gear with a pressure angle of 20, the minimum number of teeth is $z_{min} = 17$. The profile shift coefficient x becomes –

$$x = 1 - \frac{z}{z_{min}}$$

If $z > z_{min}$, then the addendum modification coefficient becomes negative. This means that without undercutting, a negative profile shift is possible.

The tooth height can be influenced by positive and negative addendum modifications. In positive addendum modification, the contact ratio becomes smaller and tooth thickness increases at the root, whereas in negative addendum modifications, the contact ratio increases which decreases the tooth thickness at the root. The tooth modification coefficients for the pinion and gear be x_1 & x_2 . There are three cases :

Case 1: The sum of addendum modification coefficients is greater than zero.

$$S_1 + S_2 > \pi m$$

Case 2: The sum of addendum modification coefficients is less than zero.

$$S_1 + S_2 < \pi m$$

Case 3: The sum of addendum modification coefficients is equal to zero.

The modified gear formulas are applied to the first and second cases.

3.2 Working pressure angle (α) :

The angle ϕ describes the thickness of the involute tooth,

$$\phi = \tan \alpha - \alpha \quad (1)$$

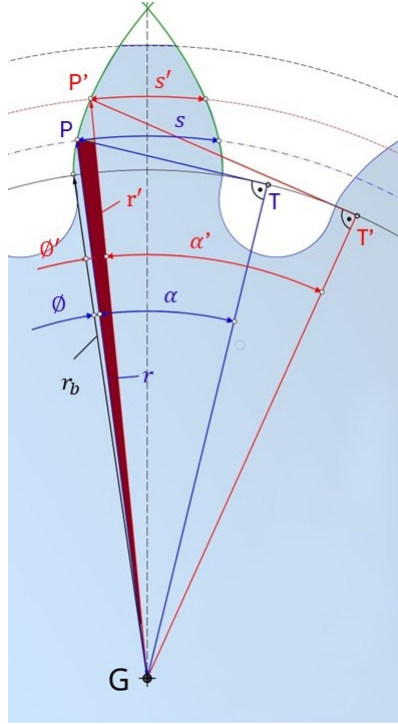


Figure 3: Calculation Of Working Pressure Angle [11]

The acute angle of the dark red triangle can be determined by the difference δ and δ' , which are in radians and describe the ratio of arc length to the length of radius,

$$\delta = \frac{s}{d} \quad , \quad \delta' = \frac{s'}{d'}$$

On the other hand, the acute angle of the yellow triangle can also be determined by ϕ & ϕ' . Thus equating the relation of δ & δ' and ϕ & ϕ' .

$$\delta' - \delta = \phi - \phi'$$

$$\frac{s'}{d'} - \frac{s}{d} = \phi - \phi'$$

$$s' = d' \left(\frac{s}{d} + \phi - \phi' \right) \quad (2)$$

Using Equation (1),

$$s' = d' \left(\frac{s}{d} + \text{inv}(\alpha) - \text{inv}(\alpha') \right) \quad (3)$$

Substituting tooth thickness formula in (3), and $m = \frac{d_o}{z}$,

$$s' = d' \left(\frac{1}{z} \left(\frac{\pi}{2} + 2x \times \tan(\alpha) \right) + \text{inv}(\alpha) - \text{inv}(\alpha') \right) \quad (4)$$

In this case, the involute function $\text{inv}(\alpha)$ refers to the operating pitch circles d'_1 or d'_2 , where the involute angle α' corresponds to the operating pressure angle.

For backlash-free meshing, the sum of respective tooth thickness s_1 and s_2 thus corresponds to the circumferential pitch p on the operating pitch circles of the gear.

$$p' = s'_1 + s'_2$$

$$p = d'_1 \left(\frac{1}{z_1} \left(\frac{\pi}{2} + 2x_1 \times \tan(\alpha) \right) + \text{inv}(\alpha) - \text{inv}(\alpha') \right) + d'_2 \left(\frac{1}{z_2} \left(\frac{\pi}{2} + 2x_2 \times \tan(\alpha) \right) + \text{inv}(\alpha) - \text{inv}(\alpha') \right) \quad (5)$$

Circular pitch: $p = \frac{\pi d}{z}$, therefore $d = \frac{zp}{\pi}$ Substituting this value in Equation (5):

$$p = \frac{z_1 p}{\pi} \left(\frac{1}{z_1} \left(\frac{\pi}{2} + 2x_1 \times \tan(\alpha) \right) + \text{inv}(\alpha) - \text{inv}(\alpha') \right) + \frac{z_2 p}{\pi} \left(\frac{1}{z_2} \left(\frac{\pi}{2} + 2x_2 \times \tan(\alpha) \right) + \text{inv}(\alpha) - \text{inv}(\alpha') \right) \quad (6)$$

Solving the above equation [10] [11]:

$$\text{inv}(\alpha) = \text{inv}(\alpha) + 2 \times \tan(\alpha) \left(\frac{x_1 + x_2}{z_1 + z_2} \right)$$

Here, x_1 & x_2 are addendum modification coefficients and z_1 & z_2 are the teeth of pinion and gear. It should be noted that a reverse function cannot be obtained because the involute function is not algebraic. The working pressure angle may be established in this situation using the iterative Newton's approach.

3.3 Base circle radius (r'_b):

[11]

$$r'_b = r \times \cos(\alpha) = \frac{zm}{2} \cos \alpha$$

Where, r'_b is the modified base radius.

3.4 Pitch circle radius (r'):

[11]

$$r' = \frac{d'}{2} = \frac{r_b}{\cos \alpha'} = \frac{r \cos(\alpha)}{\cos \alpha'} = \frac{zm \cos(\alpha)}{2 \cos \alpha'}$$

Where, r is the modified pitch circle radius.

3.5 Addendum coefficient (h'_a) & Addendum height (h'_a):

$$h'_a = 1 + x$$

$$h'_a = m \times h'^*_a$$

3.6 Dedendum coefficient (h'_f) & Dedendum height (h'_f):

$$h'^*_f = 1.25 - x$$

$$h'_f = m \times h'^*_f$$

3.7 Circular pitch (p'):

$$p' = \frac{2\pi r'}{z} = \pi m' = \frac{\pi m \cos \alpha}{\cos \alpha'}$$

Where,

$$m' = \frac{m \cos(\alpha)}{\cos \alpha'}$$

3.8 Tip circle (d'_a):

The addendum height is modified as $h_a = m + mx$, which modifies the tip circle as [11]

$$d'_a = d' + 2m + 2mx$$

3.9 Root circle (d'_f)

The dedendum is calculated using $h_f = 1.25m - mx$, the root circle is calculated as

$$d'_f = d' - 2.5m + 2mx$$

3.10 Root radius (r'_f)

$$r'_f = \frac{d_f}{2}$$

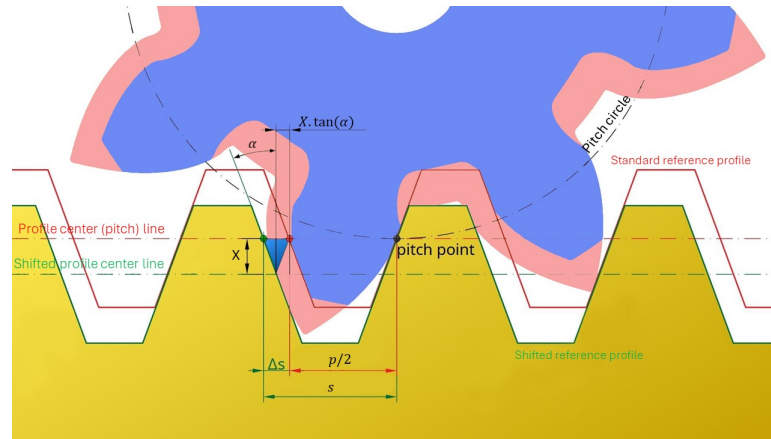


Figure 4: Calculation Of Tooth Thickness (Techscience [11])

3.11 Tooth thickness (s'):

The profile shift ' X ' is indicated by the profile shift coefficient x to the module m .

$$X = m \times x$$

Here, s is the thickness of the tooth along the circular pitch and Δs is the shift distance as shown in the fig.(3). The thickness of the tooth can be represented as

$$s' = \frac{p}{2} + \Delta s$$

$$s' = \frac{p}{2} + 2 \times X \times \tan(\alpha)$$

Since, $p = \pi m$ & $X = mx$, we get [9] :

$$s' = \frac{\pi \times m}{2} + 2mx \times \tan(\alpha)$$

$$s' = m\left(\frac{\pi}{2} + 2x \times \tan(\alpha)\right)$$

Tooth space (e'): The tooth space is calculated by [9]

$$e' = m\left(\frac{\pi}{2} - 2x \times \tan(\alpha)\right)$$

3.12 Center distance (a')

$$\begin{aligned} a' = r'_1 + r'_2 &= \frac{m'(z_1 + z_2)}{2} = \frac{z_1 + z_2}{2} \frac{m \cos(\alpha)}{\cos \alpha'} \\ &= a \frac{\cos(\alpha)}{\cos \alpha'} \end{aligned}$$

3.13 Root fillet radius (ρ_f):

[10]

The root fillet radius is the radius of the curve that connects the bottom of the gear tooth flank to the gear wheel's root.

$$\rho_f = \frac{r_T + (h_f - r_T)^2}{0.5d + (h_f - r_T)}$$

Here, ρ_f is minimum calculated root fillet radius produced by hobbing, generating rack or generating grinding and r_T is the edge radius of the generating hob or grinding wheel.

Table 1: Spur gear proportions to find the r_T value [10]

Application	Pressure angle α (in degrees)	Edge radius r_T (in mm)
General purpose	20	0.300
Extra depth for shaving	20	0.350
Aircraft, full fillet, high fatigue strength	20	0.380
Alternative high-strength design	25	0.300
Fine pitch gears	20	-

According to Stephan R [10] values for the r_T can be selected from the Table 2 The values are for 1 module. For other modules, multiply the value

by module. For the case study, general purpose value of r_T is used, which is 0.300, and should be multiplied by module 2 to get the value of the edge radius.

3.14 Length of line of action (Z')

The length of the line traced between the teeth's first point of contact and the last point of contact [12].

$$Z' = \sqrt{r_{a1}^2 - r_{b1}^2} + \sqrt{r_{a2}^2 - r_{b2}^2} - a' \sin(\alpha')$$

Where, r_{a1} and r_{a2} is the tip radius of the pinion and gear
 r_{b1} and r_{b2} is the tip radius of the pinion and gear,

3.15 Contact ratio (ε_α)

Contact ratio is the ratio of the length of the path of contact to the base pitch, and is calculated as

$$\varepsilon_\alpha = \frac{z'}{p_b}$$

Where, p_b is the base pitch, ε_α is the transverse contact ratio.

3.16 Hertzian Contact Stress (σ_c) :

The interaction of two involute teeth is comparable to the contact between two cylinders or disks. Hence, the Hertz equation can be utilized to measure contact stresses between gear teeth. The Hertzian equation for contact stress between gear teeth is as follows [5].

$$\sigma_c = \sqrt{\frac{F \left(1 + \frac{r_1}{r_2}\right)}{r_1 b \pi \left[\frac{(1-v_1)^2}{E_1} + \frac{(1-v_2)^2}{E_2} \right] \sin(\alpha)}}$$

Where, r_1 & r_2 are pitch radii of two mating gears, F is the force, v_1 & v_2 are Poisson ratio and E_1 & E_2 are the modulus of elasticity of two gears in mesh.

For the case study calculation, 100Nm moment is applied. Also, Carbon Steel (36Mn5) is used as material for both pinion and gear, since it has low

cost, toughness & good tooth hardness. Hence the value of the Poisson ratio and modulus of elasticity for pinion and gear are-

$$v_1 = v_2 = 0.3 \quad \& \quad E_1 = E_2 = 206GPa$$

4 Gear Calculation for the three cases

For generating the mathematical model of involute gear with 3D printing and displaying the effect of tooth height, one should have all the required mathematical geometry. The 3D-printed gear can be generated using the Lua script and IceSL. Firstly, the basic parameters like module, pressure angle, number of teeth, and profile shift coefficient should be defined as shown in the figure. Then find out the working pressure angle and thickness. Since Lua is a strong language script that is comparable to C, it can produce detailed gear for IceSL, a potent slicer program.

Table 2: Gear specifications

Gear specifications					
Gear Parameters	Unit	Pinion 1	Pinion 2	Pinion 3	Gear
Module	mm	2	2	2	2
Diametral pitch	mm	0.5	0.5	0.5	0.5
Number of teeth	-	14	14	14	20
Pressure angle (in degrees)	Degree	20	20	20	20
Involute angle (in radians)	Radian	0.01404	0.01404	0.01404	0.01404
Pitch circle radius	mm	14.1959	14	13.7805	20
Base circle radius	mm	13.1557	13.1557	18.7939	18.7939
Circular pitch	mm	6.2832	-	-	-
Addendum modification coefficient	-	+0.25	0	-0.25	0
Face width	mm	20	-	-	-
Tooth thickness	mm	3.5056	3.1416	2.7776	3.1416
Tooth space	mm	2.7776	3.1416	3.5056	3.1416
Addendum coefficient	-	1.25	1	0.75	1
Addendum height	mm	2.5	2	1.5	2
Dedendum coefficient	-	1	1.25	1.5	1.25
Dedendum height	mm	2	2.5	3	2.5
Addendum diameter	mm	33	32	31	44
Dedendum diameter	mm	24	23	22	35
Root radius	mm	12	11.5	11	17.5
Clearance	mm	0.47574	0.5	0.46705	-
Center distance	mm	34.4757	34	33.467	-
Operating pressure angle	Degree	22.07	20	17.32	20
Length of line of action	mm	8.4413	8.9142	9.6692	-
Contact ratio	-	1.4297	1.5098	1.6377	-
Root fillet radius	mm	0.72567	0.82704	0.95598	0.76484
Contact ratio	MPa	160.6671	161.7873	163.0704	-

5 Results & Discussion:

1. Comparing the values from Table 1, with positive addendum modification of $+0.25$, the tooth thickness, addendum height, center distance, and operating pressure angle increase, whereas, the contact ratio decreases.
2. For negative addendum modification of -0.25 , the tooth thickness, addendum height, center distance, and operating pressure angle decrease, whereas, the contact ratio increases.
3. The Hertzian contact stress is less (160.66MPa) in positive modification, as compared to 163.07MPa in negative modification as calculated in Table 2. This shows that the positive addendum modification reduces the Hertz contact stress by improving the load distribution and tooth robustness. In contrast, negative addendum modification can increase contact stress, which could be used to reduce gear size but must be carefully considered due to the increased risk of gear failure.

6 Educational Video Overview

The educational video is based on understanding the intricacies of the gear mechanics. The video will brief the history and application models of involute spur gear, followed by an introduction to basic gear concepts. Then, the video will delve into addendum modification, which is also known as profile shift. Visual aids will be used to illustrate the effect of positive and negative addendum modifications on the gear pairs, which will highlight the purpose of adjusting gear strength and load distribution. The animated simulations will be used to illustrate the comparison in contact stresses, which will compare the shifts in gear efficiency. Furthermore, the manufacturing processes and discussion on material selection and heat treatment will be highlighted. The video will conclude with a recap of the discussed points and promote more research into cutting-edge gear designs. The video aims to connect theoretical understanding to real-world application.

7 Further information

7.1 Advantages of addendum modification in the involute gear

- An undercut can be prevented by the positive profile shifting in the involute spur gears, resulting in a smooth torque transfer via the meshing gears.
- In order to maximize wear resistance, it balances the specific sliding.
- In order to maximize the resistance against bending fatigue, the profile shift also balances the bending fatigue life. For a particular profile shift value, the different analysis shows that the bending stress at the crucial section drops by 20–25% with an increase in pressure angle. There is a 25% decrease in the bending stress at the crucial section as a result of the positive shift. The application of pressure angle modification and pose shift both resulted in a 35–40% decrease in bending stress.
- Additionally, profile shift modifies the tooth thickness in addition to changing the center distance between two meshing gears by adding or subtracting teeth.

7.2 Disadvantages of addendum modification in involute gear

- Positive profile shifting on a gear causes the tooth tip to narrow and maybe become very pointed (teeth should never be entirely pointed since this increases the danger of breakage and cracking during hardening).
- An undercut may result from negative profile shifting on gears, which weakens the tooth root (part of the tooth flank is sliced off during construction.)
- For this gear, undercutting or interference between the teeth could happen if addendum modifications are not done correctly.

7.3 Areas of Application

Involute gears, spur and helical ones, are widely used in reducers, planetary gear trains, transmissions, and many other industrial applications [8]. Because of their efficient and dependable power transmission capabilities, involute spur gears are critical components in a wide range of mechanical

systems. These gears are typically used in vehicle transmissions, steering systems, and differentials, where they convert engine power to wheel motion or alter torque and speed. They are also vital in many forms of machinery, such as lathes, milling machines, and conveyor systems, which are widely used in manufacturing equipment for accurate movement control. Involute spur gears guarantee smooth functioning in household appliances like washing machines and food processors by transforming motor power into motion for tasks like spinning or mixing. Involute spur gears are also employed in power tools, robotic systems, and wind turbines, where they enhance the rotating speeds of turbine blades to efficient levels for generating electricity. Their application includes clocks and timing devices for accurate timekeeping.

8 Conclusion and Recommendations

8.1 Conclusion

The correction factors are being standardized for interchangeable gearing in current gear technology. In the past, gears were adjusted to either accomplish a preset center distance or prevent undercutting. These justifications remain true, but the positively corrected gear profile hypothesis has additional advantages. The mathematical model in this study addressed the procedures and methods of gear tooth profile modification based on the understanding of gear theory. It is now possible to modify tooth profiles consistently. Precisely designing the involute portion of the gear tooth shape is crucial in this subject. The influence of the tooth height in an involute spur gear pair can be examined by this approach and an efficient gear pair can be chosen.

Now, it is possible to make the mathematical model in Lua script using the parametric equations of the involute curve. It is possible to build the spur gear with any number of teeth and modules. The Lua script is run on the command prompt of IceSL, hence IceSL software is needed. The resulting gear can be easily printed in 3D using Additive manufacturing technologies.

8.2 Recommendations

- To avoid tip sharpening in positive modification, make sure that the addendum modification coefficient covers the entire path of contact.
- Efficient meshing can be achieved by reducing the friction loss on the tooth profile and using suitable lubrication conditions.

References

- [1] https://khkgears.net/new/images/gear_terminology/terminology.jpg. [Accessed 28-04-2024].
- [2] V Atanasiu, C Opreșan, and D Leohchi. Selection criteria of the addendum modification coefficients of spur gear pairs with smaller number of pinion teeth. In *IOP Conference Series: Materials Science and Engineering*, volume 147, page 012075. IOP Publishing, 2016.
- [3] V Suresh Babu and KM Abubacker. Development of involute profiled spur gear model with excel spreadsheet, solid works and cad technique. *Int. J. Mech. Eng*, 5:5–11, 2018.
- [4] Sándor Bodzás. Analysis of the effect of the addendum modification coefficient for contact surfaces of spur gear. *Strojnícky časopis-Journal of Mechanical Engineering*, 69(1):5–16, 2019.
- [5] Shiferaw Damtie and Daniel Tilahun. Contact stress analysis of involute spur gear by finite element method (fem). *Zede Journal*, 32:33–40, 2014.
- [6] Ali Raad Hassan. Contact stress analysis of spur gear teeth pair. *International Journal of Mechanical and Mechatronics Engineering*, 3(10):1279–1284, 2009.
- [7] Shuting Li. Effect of addendum on contact strength, bending strength and basic performance parameters of a pair of spur gears. *Mechanism and Machine Theory*, 43(12):1557–1584, 2008.
- [8] Faydor L Litvin and Alfonso Fuentes. *Gear geometry and applied theory*. Cambridge university press, 2004.
- [9] Shweta Nayak and Swetaleena Mishra. *Effects of addendum modification on root stress in involute spur gears*. PhD thesis, 2007.
- [10] Stephen P Radzevich. *Dudley’s handbook of practical gear design and manufacture*. CRC press, 2016.
- [11] Vincenzo Vullo and Vincenzo Vullo. Gears: General concepts, definitions and some basic quantities. *Gears: Volume 1: Geometric and Kinematic Design*, pages 1–38, 2020.
- [12] Jonathan P Wood and Gavin New. Design and manufacturing considerations for modified involute gearing. 2021.